

Ethan W. Rietz

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Summary

Backend and platform engineer leading the whole server side at a startup: API architecture, Azure infrastructure, MySQL, CI/CD, and the AI platform, owned from design through production operations.

Technical Skills

Languages	TypeScript, JavaScript, SQL, Go, Python, C#, Bash
Backend	Node.js, REST, GraphQL, WebSockets, RabbitMQ, Express
Data	MySQL (schema design, query optimization, administration), dbmate migrations, Typesense, Algolia
Cloud & Infrastructure	Azure (App Service, Functions, MySQL Flexible Server, API Management, Blob Storage, CDN, Service Bus), GitHub Actions, Docker, Ansible, Linux
AI / LLM	Vercel AI SDK, Anthropic and OpenAI APIs, tool use and structured outputs, prompt caching, agent frameworks, MCP
Integrations	Shopify (Admin GraphQL, App Store, OAuth, webhooks), Braintree, FedEx / UPS / EasyPost, Logicbroker, Klaviyo, Postmark

Professional

Coreleap

Jan 2023 - Now

Software Engineer, Peg and The Rush Market

Omaha retail-technology company operating The Rush Market and Peg. Formerly Rush ReCommerce.

Lead backend engineer. Owner of backend architecture, cloud infrastructure, databases, and release engineering across both product lines. Peg is an AI agent that processes ecommerce returns in real time, deciding for each item whether to refund it, swap it for an exchange or store credit, offer the customer a discount to keep it, send it back to the retailer, or resell it.

- **Founding backend engineer for Peg.** Authored roughly 90% of the primary API service, defined the layered service architecture the team still builds on, and remained its top contributor for two years.
- **Built the AI platform behind Peg's return decisions.** Skills that read customer photos, assess damage, judge resale condition, and check an item against the retailer's return policy all run through a provider-agnostic LLM client with retries, circuit breakers, and per-skill fallback models, so a vendor outage degrades the experience instead of breaking it. Prompts, model choice, and reasoning tier live in a database-driven registry, tunable without a deploy.
- **Built the resale side of the product,** where returned items still worth selling go to live auction rather than back to a warehouse. Proxy bidding with dynamic increments, auction expiry and win handling, auto-relisting, buy-it-now, auto-checkout, and penalties for winners who do not pay, with bid updates pushed over WebSockets. The winning bidder receives the item directly from the customer who returned it, so it never makes the return trip.
- **Built the returner profile and risk scoring** behind Peg's fraud prevention, which flags serial returners and blocks high-risk accounts before a return is approved.
- **Designed and shipped the public Shopify app** that retailers install to connect their store. Took it through App Store commercialization: usage-based volume pricing through the App Events API with deterministic idempotency, the mandatory privacy-compliance webhooks, an OAuth token model migration from offline to expiring online tokens, and a REST to GraphQL Admin API migration.

- **Cut Azure infrastructure spend 76%**, an annual saving on the order of a full engineering salary, through a self-scoped audit and remediation program: app-service rationalization, Windows to Linux migrations to shed licensing cost, whole-environment decommissioning, and three-year reserved capacity, while absorbing an entirely new product stack inside the reduced envelope.
- **Migrated the entire production MySQL estate** (5 hosts, 23 Azure App Service targets across 8 services) onto new Flexible Server instances with zero data loss, reclaiming 4.8 TB of storage (−77%) and carrying a MySQL 5.7 to 8.0 upgrade in the same cutover. Wrote an orchestration CLI with dry-run diffing and one-command rollback, and rehearsed the cutover against preprod first.
- **Consolidated three Node services and five private packages into one strict-mode TypeScript pnpm monorepo** with per-app change-detection CI/CD, so a change in one app builds and deploys only that app. Introduced dbmate schema migrations, a CI drift check, a house migration linter, and a Dockerized DB-backed integration test suite.
- **Integrated Logicbroker dropship end to end:** orders, inventory, product feed, shipments, returns, and invoicing for 100+ vendors with zero manual steps, plus the reusable API client package.
- **Owned the RushMarket API** behind rushmarket.com, the Corelink warehouse admin tool, and the vendor-facing returns portal: pricing and markdown engines, liquidation and multi-box processing workflows, cycle counting, and the operational reporting suite. Also the **Geographic Display Engine**, which pulls carrier ship costs and calculates profit margins across the major US metro shipping zones, and the **external API** that vendors and third-party logistics partners integrate with directly, published and secured through Azure API Management.
- **Primary reviewer and merge gate for the backend codebase.** Built the team's Jira, database, and PR-review agent tooling, plus the Ansible-managed VM fleet the engineering team's AI coding agents run on. Occasional small feature work in the Vue/Quasar frontend.

Werner Enterprises

Apr 2021 - Nov 2022

Software Intern (Oct 2021 - Nov 2022)

- Built a real-time data sync application between legacy Werner applications and the Mastery Logistics Systems Mastermind TMS, using Azure Service Buses for reliable message delivery.
- Developed mobile applications with the Ionic framework on a C#/.NET backend, with extensive unit and integration test suites in MSTest.

Tech Stack: C#, TypeScript/Angular, Azure, SQL Server, Entity Framework

Software QA Intern (Apr 2021 - Oct 2021)

- Tested APIs, UIs, and mobile devices using a variety of testing methods.

Tech Stack: Postman, Perfecto, AS400, Db2, Azure Cosmos DB, Python, JavaScript

Education

Oregon State University

Jan 2021 - Aug 2022

Bachelor of Science: Computer Science (GPA: 4.0)

Indiana University Bloomington

2019 - 2021

Ph.D. Student: Physical Chemistry and Scientific Computing (GPA: 3.76)

Iowa State University

2013 - 2017

Bachelor of Science: Biochemistry (GPA: 3.51)